KUKU FM AI GENERALIST ASSIGNMENT — SOLUTION DOCUMENT

Candidate: Zeeshan Mehmood | **Problem 1:** Find the Promos (Ads) | **Show:** *I Married a Broken Soldier. He was a Duke* (50 Episodes)

1. Core Problem & Division of Labor (Code vs. LLM vs. Human)

Today, human editors watch every microdrama episode to manually locate 20–60s ad cuts for TikTok, Reels, and YouTube Shorts. A naive single-pass LLM prompt fails: it hallucinates non-existent timestamps, cuts mid-sentence, and suffers from self-grading bias. Our architecture enforces strict **separation of concerns**:

• **Deterministic Code:** 0.01s script precision regex parsing (3,723 elements parsed in under a second), directional boundary snapping, duration boundaries (20–60s), ad-policy regex, and weighted composite math.

• **LLM Judgment:** Subjective moment identification, 0–3s hook recognition, and adversarial critique. *The LLM never writes timestamps*; it selects stable script-element IDs ([01.014]), which code resolves to verbatim source boundaries.

• **Human Editor in the Loop:** Reviews the ranked 15-promo shortlist, verifies cut points (±1–2s fine-tuning for actor breath/music), and selects creative angles for weekly ad rotation.

2. 5-Pass Production Pipeline & Measured Execution Cost

Pass	Stage	Engine / Mechanism	Responsibility & Measured Performance
0	Script Parsing	Deterministic Regex	Parses 50 episodes (3,723 elements, 1,483 dialogue lines) in under a second.
1	Dramatic Moments	gemini-flash-lite-latest	Discovers high-arousal beats (slaps, proposals, reveals) via ID-anchoring.
2	Promo Construction	Anchor & Expand (Code)	Builds 20–60s single & cross-episode candidates; snaps starts/ends to script boundaries.
3	Blind Funnel Scorer	Separate Blind Prompt (Same Family)	Scorer sees only clip text (never generator claims); weighted formula computed in code.
4	Adversarial Auditor	Dual-Layer Gate + Auto-Trim	Layer 1 regex (ad-policy, quotes) + Layer 2 LLM critique screen for context & auto-trims boundaries.

Measured Pipeline Metrics (50 Episodes): Model: `gemini-flash-lite-latest` | Real API Calls: **266** | Total Cost: **$0.0934** ($0.0019/ep) | Wall-Clock Time: **11.2 min (673s, initial uncached run)** | Validation Pass Rate: **82.6%** | Coverage: **15/50**.

Heuristic Fallback Disclosure: Moment discovery used heuristic fallback on Ep 24, Ep 45 (Gemini returned 0 beats) (shortlist: PR-042, PR-080); candidates passed full LLM construction and blind scoring.

3. Top Ranked Promo Candidates (Programmatically Extracted with Segment Timestamps)

Rank	Promo ID	Score	Episodes & Segment Timestamps	Dur	Angle	Hook & Cliffhanger Summary
#1	PR-018	8.75	Ep10 00:02.10-00:38.90	36.8s	Survival	**Hook:** Alexander abruptly bursts through the wooden cottage door, clutching his blood-soaked chest while Juliana gasps in sheer terror. **Cliff:** Julianna gasps Dear God as Alexander strips off his shirt, exposing a brutal web of jagged, fresh battle scars across his entire torso.
#2	PR-080	8.50	Ep45 00:51.80-01:10.50 + Ep45 01:26.50-01:53.00	45.2s	Betrayal	**Hook:** Alexander stands right up to Catherine, towering over her with righteous fury and declaring: "You poisoned my mother. You tortured my wife." **Cliff:** The High Priest scowls directly at Juliana with unyielding authority, proclaiming: "You are not Alexander Ashburton's lawful wife."
#3	PR-008	8.20	Ep5 00:54.00-01:14.50 + Ep5 01:31.00-01:40.80	30.3s	Betrayal	**Hook:** Lady Ashcroft violently wrenches the treasured heirloom ledger out of Juliana's grip and flings it disdainfully into the rain-soaked courtyard. **Cliff:** Gerald stands high on the stone steps, mocking Juliana as a beggar's wife, while her new husband delivers a lethally cold warning: 'Choose your next words carefully.'
#4	PR-029	8.20	Ep15 00:44.20-01:26.00	41.8s	Romance	**Hook:** The Thug with Cap violently shoves Juliana back with both hands; she cries out as she loses her balance and crashes hard onto the wet stone path. **Cliff:** Alexander steps directly between Juliana and the thugs with deadly intent, and the Thug with Cap challenges him: What? You want to fight us, veteran?

4. Benchmark Comparison vs. Baselines (Multi-Judge Blind Scoring)

Strategy (Baselines N=14–15)	Judge 1 (gemini-flash-lite-latest)	Judge 2 (gemini-3.5-flash-lite)	Empirical Analysis of Performance
Random Baseline (20-60s, N=14)	6.27	6.23	Truly random windows snapped to elements. Often starts on dead air.
Last-30s Baseline (N=14)	6.25	6.28	Episode endings with terminal cliffhangers (human scored 8.1).
Rule-Based Baseline (N=15)	6.00	6.07	Max density of emotional shockwords ('slap', 'kill'). Highest keyword intensity.
Pipeline: All Candidates (N=92)	6.40	—	Raw average across all 92 candidate promos generated across 50 episodes.
Pipeline: Best Per Episode (N=50)	6.88	—	Top scoring single candidate for each of the 50 episodes.
Pipeline Final Shortlist (N=15)	7.89	7.57	Top 15 ranked promos balancing hook (30%), cliffhanger (25%), coherence (20%).

Second-Judge Rescoring (gemini-3.5-flash-lite): Under the second judge, same model family (`gemini-3.5-flash-lite`), the pipeline shortlist (7.57) outperforms all baselines (+1.29 vs last-30s, +1.34 vs random, +1.50 vs rule-based; top-15 shortlist vs unselected baselines; biased upward) with cross-judge rank correlation ρ = 0.848.
Selection-Bias Disclosure: Note: the 7.89 shortlist is the top 15 of 92 candidates, compared with unselected baselines, so it is biased upward. The fairer comparisons are all candidates (6.40) and best-per-episode (6.88) against baselines of 6.0–6.3. The pipeline's gain over baselines is modest (about 0.1–0.6). In the single-rater blind human test, human raters preferred last-30s endings (8.1 vs 7.9), likely because screenwriters design episode finales as cliffhangers (an untested hypothesis; at N=25 the 0.2 gap is within noise).

5. Single-Rater Blind Human Evaluation & Pairwise Comparison

• **Absolute Ratings (N=25):** Shuffled blind rating test yielded Spearman correlation ρ **= 0.313** between human and LLM (weak positive, non-significant at N=25; requires multi-editor panel).
• **Pairwise Blind Comparison:** Pairwise comparison not completed in this submission. *Caveat:* single rater, small N.

6. Honest Quality Audit: Real Flagged Examples & Auto-Trim Fixes

• **Ad-Policy Tiering (2 hard trimmed, 9 hard excluded, 15 soft kept flagged):** Hard hits (suicide, hanging) are auto-trimmed if duration ≥20s or excluded from shortlist; soft hits (poison, witchcraft) remain flagged for editor review.
• **Hallucinated Claims (8 flags):** Caught claims quoting lines outside clip: **PR-009** (*'Then keep it...'*) and **PR-005** (*'Look at you...'*).
• **Delayed Hook Beat (8 flags): PR-006** claimed a warning hook, but dialogue began at [+03.9s].
• **Resolved Cliffhanger (2 flags): PR-020** and **PR-053** were flagged; both were excluded from the shortlist.

7. Stress Testing & Empirical Robustness Audits

Test Suite	Result	Empirical Findings from Stress Harness (stress_results.json)
Parser Robustness	9/9 PASS	Handles empty, header-only (raises ValueError), comma decimals, 1-3 decimals without silent 0 elements.
Timestamp Integrity	107/107 Valid	0 boundary violations across all candidate and ranked promos (1 micro-segment warning <3s).
Adversarial Audit	3/4 Caught	Catches slow hooks, hallucinated quotes, and ad policy; resolved cliffhanger unflagged (rule checker miss).
Rerun Stability	2/3 Stable	Top candidate segment IoU >= 0.5 across un-cached re-runs on Ep 1-3.
Judge Discrimination	PASS	Mean spread 0.27 across repeats; discriminates good vs flat promos with a +1.24 score gap.

8. Production Scaling & Stated Assumptions

• **Production Scaling:** Validated across 50 episodes with real Gemini calls. Catalog processed in 11.2 min (673s, initial uncached run) for $0.0934 ($0.0019/ep). A 1,000-episode catalog scales to ~$1.87 in API costs.
• **Tools & Where They Failed:** Python 3.12, Google GenAI SDK, Pydantic v2, ReportLab, Streamlit. Gemini initially hallucinated floating-point timestamps in natural language, motivating discrete element-ID anchoring (`clips.py`).
• **Assumptions:** Assumes human editors spend ~15-20h/50 eps logging cuts (unverified estimate); conversion weights are untested priors; screenplay timestamps provide 0.01s precision with ~1-2s human trim adjustment for breath/music.